Developing a Muscle Tissue Construct Using 3D Bioprinting

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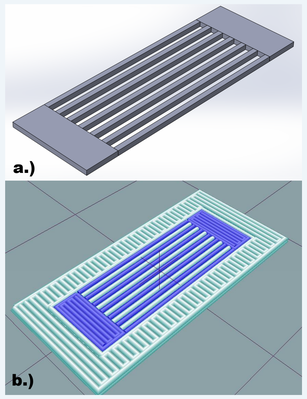
Introduction

- Tissue engineering aims to use biomaterials, stem cells, and other biomolecules to replace or repair damaged human tissue and organs.
- Bioprinting is a tissue engineering approach that uses additive manufacturing to build biomimetic structures in a layer-by-layer fashion.
- Creation of muscle tissue is a major goal of tissue engineering. Bioengineered skeletal and smooth muscle could be used to repair volumetric muscle loss and gastrointestinal (GI), vascular, or urinary tract damage.
- Proper muscle contraction depends on cell alignment, which has been a limitation for creating functional muscle[1].
- Engineered tissue cannot support cells deeper than ~200µm into a construct due to oxygen diffusion limits[2].
- Bioprinted skeletal muscle tissue constructs have been successfully produced that have aligned muscle cells and features small enough to allow diffusion to cells[3,4,5].
- Few attempts at bioprinting GI smooth muscle have been made.

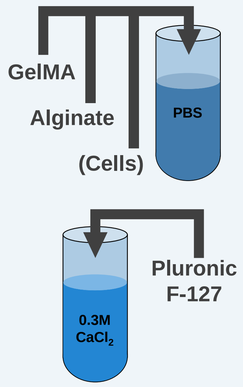
Objective

To create a bioprinted construct that has features conducive to cell alignment and viability for use in muscle bioengineering research.

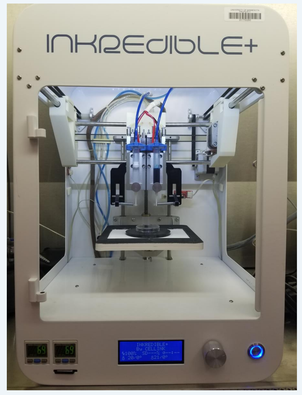
Materials and Methods



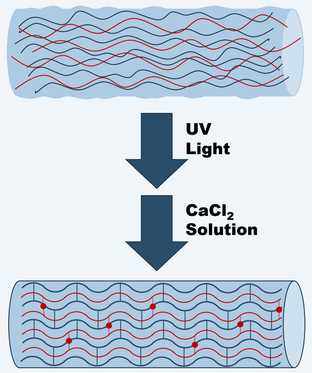
Computer model of the construct without support structure (a.) and visualization of g-code with GelMA/alginate hydrogel in dark blue and Pluronic-F127 in light blue (b.)



Representation of mixing the GelMA/alginate blended hydrogel (top) and the Pluronic F-127 support hydrogel (bottom) used in this project.



The Inkredible+ bioprinter used in this project.



Representation of hydrogel crosslinking. Exposure to UV light and Ca^{2+} ions causes individual GelMA or alginate molecules to bond, thus solidifying the hydrogel's structure.

File Creation

CAD Models Designed and Converted to G-code

Bioprinters are controlled by a numerical programming language called g-code.

Computer models were created using computer-aided design (CAD) software. Models were converted to g-code using 3D printing slicing software.

SolidWorks, Repetier, and Slic3r were used for computer modeling, g-code editing, and model slicing.

Materials Preparation

Hydrogels Formulated

Biocompatible materials that can take the form of hydrogels are used as bioink.

A hybrid hydrogel composed of 5% (w/v) gelatin-methacryate (GelMA), 2% alginate, and 0.5% LAP photoinitiator was created as a low-viscocity bioink.

A secondary hydrogel of 30% (w/v) Pluronic F-127 dissolved in 0.3M $CaCl_2$ was used to print a support structure for the primary bioink.

Bioprinting

Constructs Created

Bioinks are deposited in the pattern dictated by the g-code file to make a bioprinted construct.

The Inkredible+ bioprinter was used to create constructs. This printer uses pneumatic extrusion to control placement of bioink.

The Pluronic F-127 support structure was printed first to contain the GelMA/alginate hydrogel before crosslinking.

Post-Processing

Hydrogels Crosslinked and Support Gels Removed

Bioprinted constructs are physically and chemically treated to enhance their structural integrity.

After printing, constructs were exposed to 405nm UV light to crosslink the GelMA component of the hydrogel.

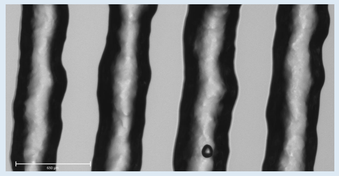
A cold 0.3M $CaCl_2$ solution was then used to wash away the Pluronic F-127 support gel and complete crosslinking of alginate.

Preliminary Results

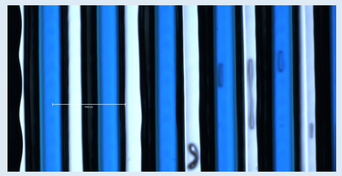
A GelMA/alginate hybrid bioink was developed that successfully crosslinked upon exposure to UV light and $CaCl_2$.

A Pluronic F-127/$CaCl_2$ support gel was successfully used to contain the low viscosity GelMA/alginate bioink. The alginate component of the bioink crosslinked upon contact with the support gel, allowing higher printing resolution of the bioink.

Attempts to bioprint constructs with features smaller than 400µm in diameter have not yet been successful.



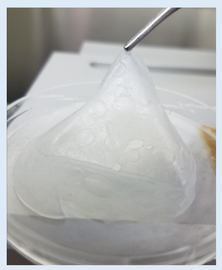
Micrograph of a bioprinted GelMA only hydrogel.

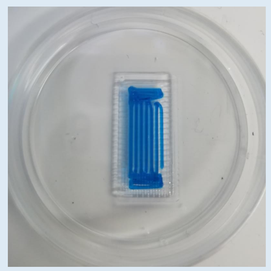


Micrograph of a bioprinted Pluronic F-127 hydrogel.

(Left) A sheet of GelMA/alginate hydrogel that has been crosslinked with UV light and $CaCl_2$ and incubated at 37° C for 4 days in PBS.

(Right) A mock print of the muscle tissue construct file created using two extruders containing Pluronic F-127, one with blue dye and one without.

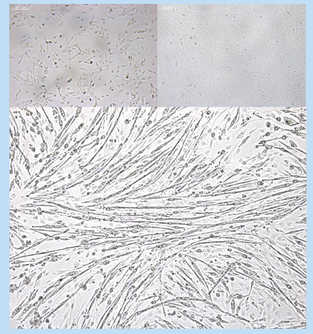




Future Research

In this study, progress was made toward creating a bioprinted construct that could serve as a muscle tissue model. Future work will explore further characterization of biocompatibility, cell alignment capabilities, and mechanical properties of the construct.

- Biocompatibility of the GelMA/alginate hydrogel will be assessed by adding mouse skeletal myoblasts to the gel prior to printing and measuring cell viability.
- Cell alignment will be assessed using image analysis software on microscope images of the construct.
- Mechanical strength will be assessed by measuring the elastic modulus of the construct using an Instron planar biaxial soft tissue testing system.



Mouse myoblasts in monolayer culture after 1 day (top left), 9 days (top right) and 14 days (bottom). Myoblasts begin to fuse and form myotubes after 14 days in monolayer culture.

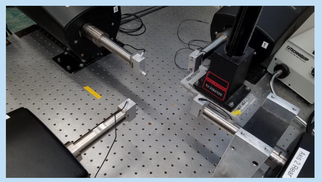


Image of the Instron soft tissue testing system that will be used to measure mechanical properties of the construct.

If the construct exhibits proper biocompatibility, cellular alignment, and mechanical strength it could be scaled-up for further skeletal muscle tissue engineering research.

Using a similar construct with smooth muscle cells for application in engineering of GI organs, especially the esophagus, is an area of particular interest.

Acknowledgements

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References

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